

PRAVEEN RAJ A

Data Scientist | Labor Market Analytics • NLP • SciPy • SQL • Machine Learning

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PROFESSIONAL SUMMARY

Data Scientist with MCA in Data Science and 2 years of graduate-level data science project experience across large dataset processing, NLP, machine learning, statistical analysis, forecasting, and deployed analytics applications. Strong in Python, SQL/DuckDB, Pandas, scikit-learn, SciPy, Streamlit, Plotly, skill extraction, model evaluation, and business insight visualization. Built a Lightcast-style labor market intelligence platform aligned with workforce analytics, future-of-work insights, and stakeholder recommendations.

TECHNICAL SKILLS

Programming / Analytics: Python, SQL, DuckDB, Pandas, NumPy, SciPy, data cleaning, statistical analysis, large dataset processing

Machine Learning / Statistics: Classification, Regression, Forecasting, Logistic Regression, Random Forest, XGBoost, SVM, Isolation Forest, K-Means, model evaluation, hypothesis testing

NLP / Data Products: Natural Language Processing (NLP), skill extraction, information extraction, TF-IDF, semantic search, RAG, Sentence Transformers, LLM APIs

Visualization / Tools: Streamlit, Plotly, Matplotlib, Seaborn, SHAP, GitHub, Jupyter Notebook, VS Code, Excel

PROJECTS

1. LaborIQ — Labor Market Intelligence & Skills Demand Analytics Platform

Links: [GitHub Project](#) | [Live App](#)

Tech: Python, SQL, DuckDB, Pandas, NumPy, SciPy, scikit-learn, NLP, TF-IDF, Logistic Regression, Ridge Regression, Plotly, Streamlit, Adzuna API

Business value: Converts raw job postings into structured workforce intelligence, future-of-work insights, skill demand trends, and stakeholder-ready recommendations.

- Built and deployed a Lightcast-style labor market intelligence platform to clean, organize, analyze, and visualize job-posting data for role demand, skill demand, salary trends, occupation mapping, and workforce planning.
- Implemented live Adzuna API ingestion with multi-query fetching, caching, deduplication, source tracking, data-quality checks, and dashboard refresh workflow; processed 1.5K+ mixed-source job postings for demo analytics.
- Developed an NLP skill extraction pipeline using a controlled skill taxonomy and alias matching to identify standardized skills such as Python, SQL, NLP, Machine Learning, SciPy, APIs, LLM, RAG, Excel, and Data Visualization from job descriptions.
- Built an occupation mapping and ML role classification layer using TF-IDF features and Logistic Regression to classify postings into role families such as Data Scientist, Data Analyst, Data Engineer, AI Engineer, ML Engineer, NLP Engineer, Business Analyst, and Research Analyst.
- Added SciPy-based statistical validation, salary modeling, skill-demand forecasting, role-skill similarity, candidate skill-gap recommendations, and interactive Streamlit dashboards for executive overview, labor analytics, job explorer, and model/data-quality metrics.

2. Enterprise Document Intelligence Platform

Links: [GitHub Project](#) | [Live App](#)

Tech: Python, Streamlit, NLP, RAG, Sentence Transformers, TF-IDF, PyMuPDF, scikit-learn, Groq API

Business value: Reduces manual document review by combining retrieval, extraction, clause analysis, risk scoring, summaries, and voice interaction.

- Built a deployed document intelligence assistant for PDF upload, document Q&A, entity extraction, clause analysis, risk review, executive summaries, and voice interaction.
- Implemented PDF extraction, text cleaning, chunking, TF-IDF search, and semantic retrieval using all-MiniLM-L6-v2 embeddings.
- Created information extraction for dates, amounts, percentages, emails, and business terms, plus clause classification for payment, termination, confidentiality, liability, warranty, renewal, and governing law and Integrated LLM-based answers/summaries using retrieved evidence and transparent keyword risk scoring into Low, Medium, and High Risk.

3. Retail Demand Forecasting & Anomaly Intelligence System

Links: [GitHub Project](#) | [Live App](#)

Tech: Python, SQL, DuckDB, Pandas, XGBoost, SVM, Isolation Forest, K-Means, SHAP, TensorFlow/Keras, Streamlit

Business value: Turns transaction data into forecasts, anomaly alerts, customer segments, product patterns, and business recommendations.

- Developed a retail analytics dashboard for KPI tracking, demand forecasting, anomaly detection, customer segmentation, association-rule mining, SHAP explainability, and business recommendations.
- Used DuckDB SQL for revenue, product, customer, and country analytics; engineered time-based features, applied time-based train-test split to reduce future data leakage, and implemented XGBoost forecasting, SVM high-demand classification, Isolation Forest anomaly detection, and K-Means RFM segmentation to convert model outputs into business-friendly insights for demand planning, customer targeting, anomaly monitoring, and revenue optimization.

EDUCATION

Master of Computer Applications (MCA) - Data Science | Jain University, Bangalore | 2023 - 2025 | CGPA: 9.55

Bachelor of Computer Applications (BCA) | Vels University, Chennai | 2020 - 2023 | CGPA: 7.71